Patrick Kariuki

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Education

Johns Hopkins University — Baltimore, MD
Bachelor of Science in Computer Science — *August 2022 – May 2026*

- **Relevant Coursework:** Data Structures, Algorithms, Full-Stack JavaScript, Artificial Intelligence, Computer Systems, Object-Oriented Software Engineering, Databases, Web Security, Operating Systems, Deep Learning.

Experience

Course Assistant — August 2024 – May 2026
Johns Hopkins University — *Baltimore, MD*

- Evaluated and provided technical feedback on complex algorithm and data structure implementations for 100+ students across Data Structures, Systems Programming (C/C++), and AI-Enabled Software Engineering courses.
- Led 6-person teams as technical PM for AI-Enabled Software Engineering, advising on LLM integration, prompt engineering, RAG pipelines, and deployment architecture, shipping 3 full-stack AI applications.

Tech Developer Intern — May 2024 – August 2024
Sponsors for Educational Opportunity (SEO) — *New York, NY*

- Built StarSight, a full-stack stargazing prediction platform integrating real-time weather data, light pollution metrics, and astronomical databases to provide location-based visibility forecasts for 50 beta users across 3 states.
- Engineered Flask REST API with secure user authentication and PostgreSQL database; implemented Redis caching layer reducing API response time from 800ms to 200ms for repeated location queries.
- Trained constellation recognition model using Principal Component Analysis for feature extraction on 10,000 labeled astronomical images, achieving 85% accuracy; deployed via Docker with Ansible-managed CI/CD pipeline.

Projects

Applications Tracker | *React, Convex, TanStack, Vercel AI SDK* — June 2026 – Present

- Engineered AI email-parsing pipeline using scheduled Convex functions and Vercel AI SDK to classify incoming application emails and auto-update statuses, with confidence-scored manual review for low-certainty cases.

SDB | *C++, Docker, Linux* — March 2026 – May 2026

- Built an x86-64 debugger in C++ supporting breakpoints, stack unwinding, and source-level stepping via ptrace syscalls and DWARF debug info parsing.

2D Game Engine | *C++, Raylib, OpenGL, Lua* — January 2026 – May 2026

- Architected a 2D game engine in C++ from scratch with an entity-component-system, broadphase collision detection (sort-and-sweep), layered rendering via OpenGL, and custom memory allocators; shipped a complete tower defense game as the capstone demo.

GoCI | *Go, Git* — January 2026 – February 2026

- Built a CLI continuous integration tool in Go automating build, test, and lint pipelines with git hook integration; used as the deployment workflow for all personal projects.

Leadership

President, Software Engineering Club — May 2025 – May 2026

- Secured $500 in corporate sponsorship and organized 5+ technical workshops on career prep, interview skills, and AI-enabled software engineering for 75+ student members.
- Founded HopBuilds, a project-based learning initiative where students build real applications for academic credit; led development of Hopkins Lost and Found platform deployed campus-wide.
- Grew active membership 40% through hands-on coding events and industry networking sessions.

Technical Skills

Programming: Python, C/C++, Java, JavaScript/TypeScript, Go, HTML/CSS
Frameworks: React, Node.js, Flask, Django, FastAPI, NumPy, scikit-learn, PyTorch
Databases: PostgreSQL, MongoDB, Firebase, SQLite, DynamoDB
Developer Tools: Git, Docker, Ansible, Cursor, Linux